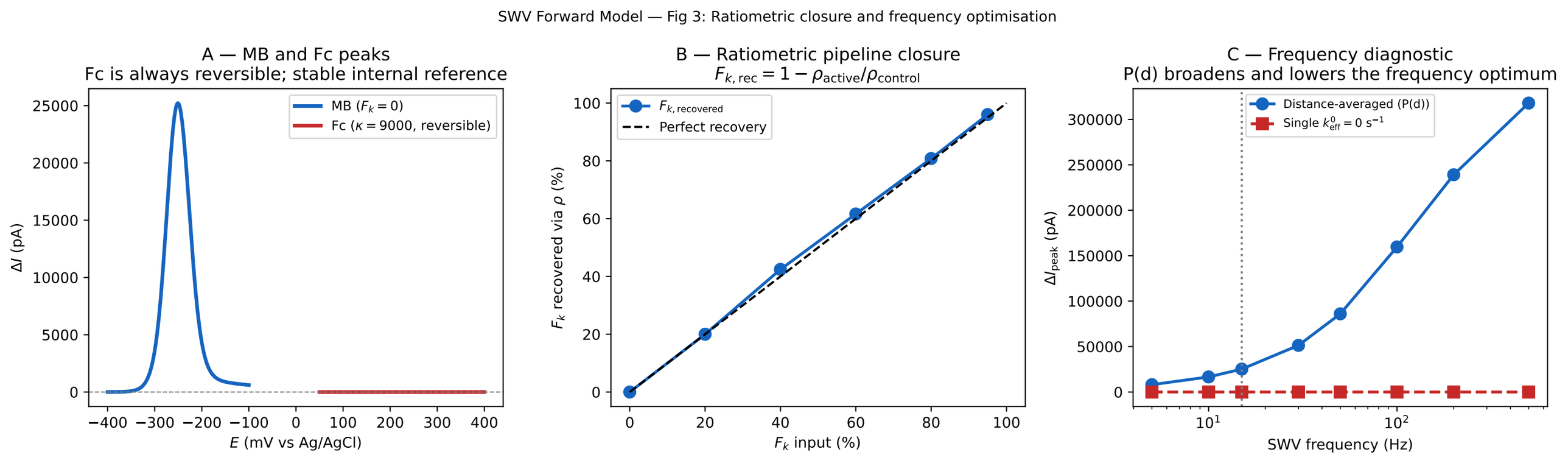
SWV Forward Model — Fig 3: Ratiometric closure and frequency optimisation
A — MB and Fc peaks
Fc is always reversible; stable internal reference
MB ($F_k = 0$)
Fc ($\kappa = 9000$, reversible)
ΔI (pA)
E (mV vs Ag/AgCl)
B — Ratiometric pipeline closure
$F_{k,\mathrm{rec}} = 1 - \rho_{\mathrm{active}}/\rho_{\mathrm{control}}$
$F_{k,\mathrm{recovered}}$
Perfect recovery
F_k recovered via ρ (%)
F_k input (%)
C — Frequency diagnostic
P(d) broadens and lowers the frequency optimum
Distance-averaged (P(d))
Single $k^0_{\mathrm{eff}} = 0\ \mathrm{s}^{-1}$
ΔI_{peak} (pA)
SWV frequency (Hz)